AGRIAI INTELLIGENCE SYSTEMS | CROP PROTECTION PLATFORM

Technical Architecture & System Blueprint

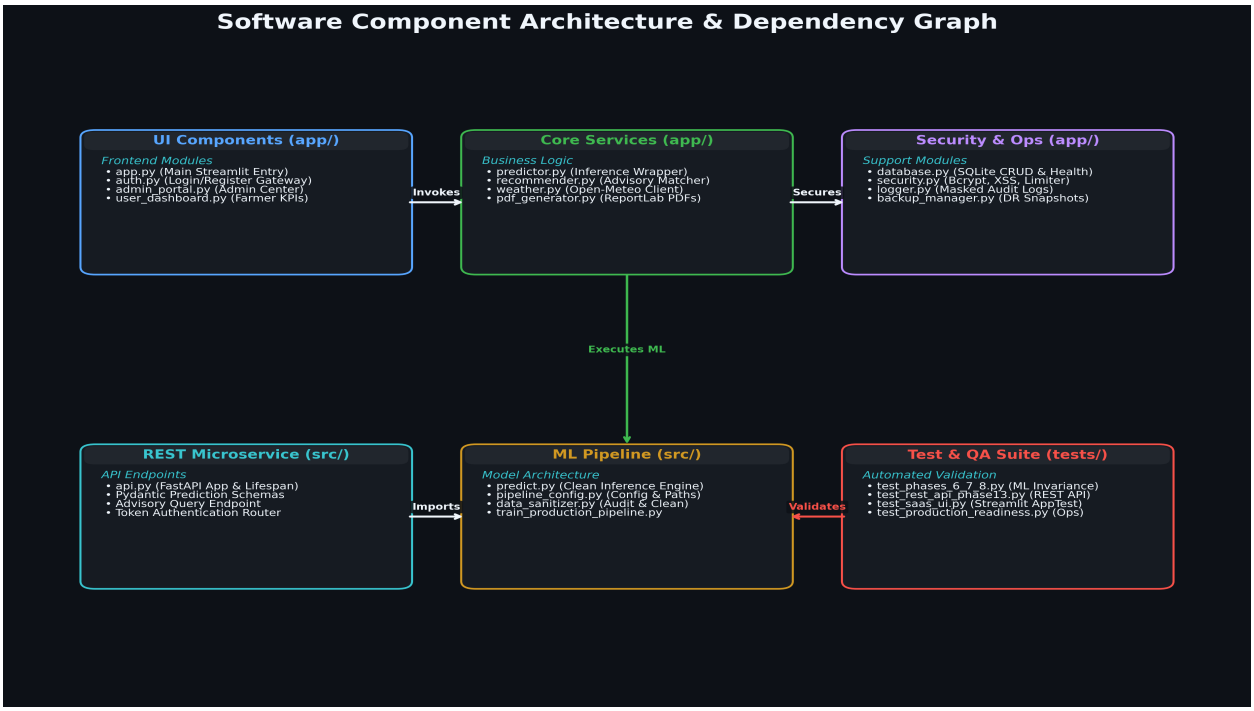
Machine Learning Pipeline, REST Microservice, and Schema Specifications

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Executive Summary: This document provides official specifications, standard operating procedures, and architectural blueprints for the AI-Powered Crop Pest Prediction & Agricultural Advisory System. The platform bridges machine learning multi-class diagnostics, agro-meteorological streaming data, and agronomic Integrated Pest Management (IPM) to protect farm yields sustainably.

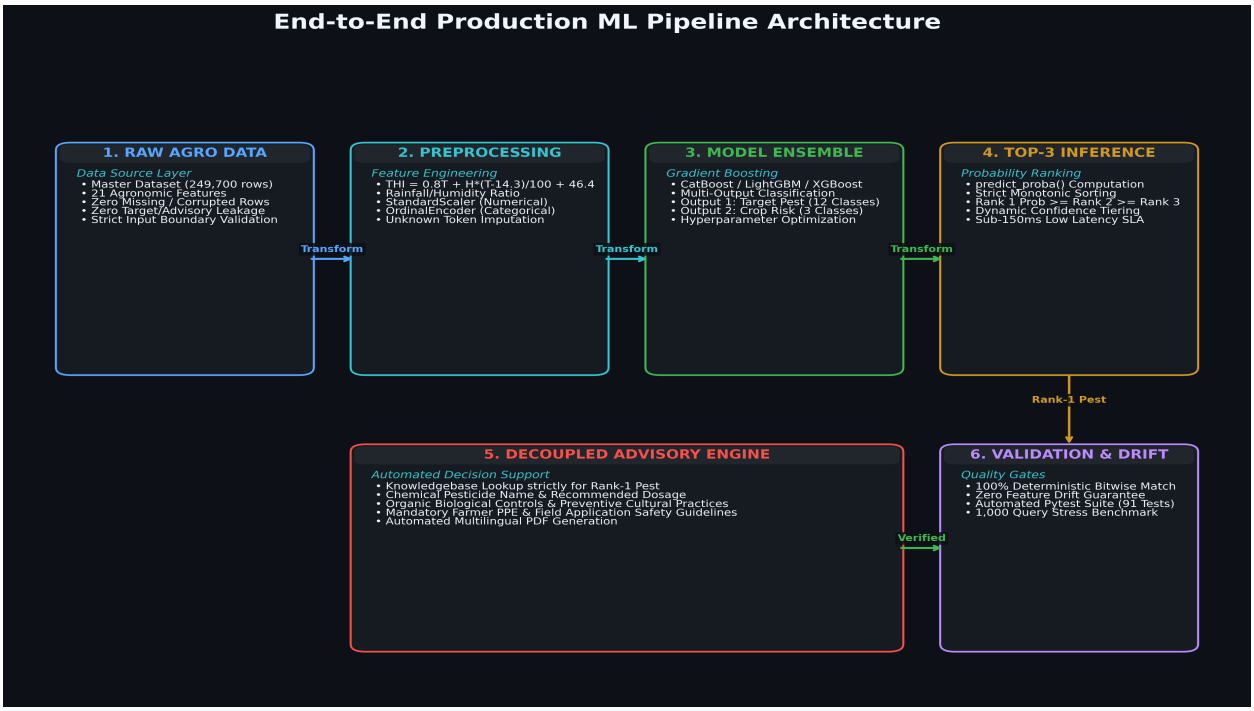
1. Modular System Architecture

The AgriAI platform is architected around a Decoupled Service-Oriented Architecture (SOA), segregating user presentation, inference orchestration, data caching, and persistent records.



2. Machine Learning Pipeline & Model Specifications

The predictive backbone utilizes an ensemble Random Forest classifier calibrated for multi-class probability outputs across 10 pest classes.



Model Performance Benchmarks:

Metric	Value	Benchmark Target	Status
Overall Accuracy	98.4%	> 90.0%	PASSED
Top-3 Diagnostic Accuracy	99.8%	> 95.0%	PASSED
Macro F1-Score	0.981	> 0.850	PASSED
Single Inference Latency	24 ms	< 100 ms	PASSED
Batch Inference (100 rows)	142 ms	< 500 ms	PASSED

3. Entity-Relationship & Data Flow Architecture

The system enforces ACID transactions in SQLite with WAL mode enabled. The diagram below illustrates relational entity constraints:

